BILANGAN BINER

LITERASI DIGITAL

Dikerjakan oleh :

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PRODI TEKNIK INFORMATIKA
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A. Konversi Biner ke Decimal

1. 3 bit (Biner 3 Digit)

000 = $(0*2^0) + (0*2^1) + (0*2^2) = 0 + 0 + 0 = 0$
001 = $(1*2^0) + (0*2^1) + (0*2^2) = 1 + 0 + 0 = 1$
010 =
011 =
100 =
101 =
110 =
111 =

2. 4 bit (Biner 4 Digit)

0000 =
0001 =
0010 =
0011 =
0100 =
0101 =
0110 =
0111 =
1000 =
1001 =
1010 =
1011 = $(1*2^0) + (1*2^1) + (0*2^2) + (1*2^3) = 1 + 2 + 0 + 8 = 11$
1100 =
1101 =
1110 =
1111 =

3. 5 bit (Biner 5 Digit)

00000 =
00001 =
00010 =
00011 =
00100 =
00101 =
00110 =
00111 =
01000 =
01001 =
01010 =
01011 =
01100 =
01101 =
01110 =
01111 =

10000 =
10001 =
10010 =
10011 =
10100 =
10101 = $(1*2^0) + (0*2^1) + (1*2^2) + (0*2^3) + (1*2^4) = 1 + 0 + 4 + 16 = 21$
10110 =
10111 =
11000 =
11001 =
11010 =
11011 =
11100 =
11101 =
11110 =
11111 =

4. 6 bit (Biner 6 Digit)

000 000 =
000 001 =
000 010 =
000 011 =
000 100 =
000 101 =
000 110 =
000 111 =
001 000 =
001 001 =
001 010 =
001 011 =
001 100 =
001 101 =
001 110 =
001 111 =
010 000 =
010 001 =
010 010 =
010 011 =
010 100 =
010 101 =
010 110 =
010 111 =
011 000 =
011 001 =
011 010 =

011 011 =
011 100 =
011 101 =
011 110 =
011 111 =
100 000 =
100 001 =
100 010 =
100 011 =
100 100 =
100 101 =
100 110 =
100 111 =
101 000 =
101 001 =
101 010 =
101 011 =
101 100 =
101 101 =
101 110 =
101 111 = $(1*2^0) + (1*2^1) + (1*2^2) + (1*2^3) + (0*2^4) + (1*2^5) = 1 + 2 + 4 + 8 + 0 + 32 = 47$
110 000 =
110 001 =
110 010 =
110 011 =
110 100 =
110 101 =
110 110 =
110 111 =
111 000 =
111 001 =
111 010 =
111 011 =
111 100 =
111 101 =
111 110 =
111 111 =

B. Makna Hexadecimal Untuk Kita (Decimal)

1. AD1 = $(10*16^2) + (14*16^1) + (1*16^0)$ = (10*256) + (14*16) + (1*1) = (2560) + (224) + (1) = **<u>2758</u>**
2. 3A6A5 = $(3*16^4) + (10*16^3) + (6*16^2) + (10*16^1) + (5*16^0)$ = (3*65536) + (10*4096) + (6*256) + (10*16) + (5*1) = (196.608) + (40.960) + (1.536) + (160) + (5) = **<u>239.269</u>**

C. Makna Hexadecimal Untuk Komputer

1. BA6A5 = 1011 1010 0110 1010 1001
2. DE5000 = 1110 1111 1001 0000 0000 0000
3. 7A7A176 = 0111 1010 0111 1010 0001 0111 0110
4. 547121A = 0101 0100 0111 0001 0010 0001 1010

D. Tuliskan "I Love U" dengan ASCII

E. Latihan

a) Konversikan bilangan desimal di bawah ini ke dalam bilangan biner
   - 123.410 = $(1\ 1110\ 0010\ 0001\ 0010)_2$
   - 567.010 = $(1000\ 1010\ 0110\ 1110\ 0010)_2$
   - 232.110 = $(11\ 1000\ 1010\ 1010\ 1110)_2$
   - 286.871 = $(100\ 0110\ 0000\ 1001\ 0111)_2$

b) Hasil biner pada bagian A, item a, b, c konversikan ke Hexa
   - 123.410 = 1E212
   - 567.010 = 8A6E2
   - 232.110 = 46097